Simulation Arena

Margin Protection & Floor Setting: Stress-Testing Cost Volatility

With no price escalator there is a 18.4% chance (about 2 in 10) this contract breaks your 12% margin floor by Year 3 - a 2.5% annual escalator is what brings that back inside your 10% tolerance.

Seed 42 | Run run_92ea2ef234e6 | 1,000 scenarios tested | Latin Hypercube Sampling of multi-year… | 21 Sep 2026, 23:38

Same seed, same numbers - this run can be reproduced exactly.

How this works

We split your cost to deliver into its moving parts and gave each one - plus the currency - a range of yearly moves, from the small ones to the nasty ones. Instead of picking randomly, we walked 1,000 complete 3-year cash flows through those ranges on an even grid, so every slice of every range gets tested and nothing important is missed. Each run asks one question: at the end of each year, is the margin still above your floor? We then repeat the whole thing at every escalator from 0% upwards until the chance of breaking the floor falls below the 10% you said you could live with. Costs are modelled with a 'bad news is more likely than good news' curve, so surprise increases weigh more than surprise windfalls.

The numbers that matter

CHANCE OF BREAKING THE 12% FLOOR BY YEAR 3
18.4%
The odds that the Year 3 margin comes in below your minimum - the final year is where cost increases have compounded the longest.
vs your tolerance is 10.0%

CHANCE OF BREAKING IT AT ANY POINT
18.5%
The odds that at least one of the years comes in below the floor.

BREAKEVEN ANNUAL ESCALATOR
2.5%
A 2.5% annual escalator brings the risk inside your 10% tolerance.
vs you are asking for 0.0%

EXPECTED MARGIN IN YEAR 3 (THE WORST YEAR)
20.3%
Year 3 is where the floor is most at risk.
vs you signed at 31.0%

MARGIN AT RISK, IN TODAY'S MONEY
$961k
The profit you planned to book minus the profit this simulation says you will actually book, in today's money.
vs planned $5.96M

CHANCE THE CONTRACT LOSES MONEY IN A YEAR
2.0%
How often a year comes in below zero margin, not just below the floor.

CHANCE OF BREAKING THE FLOOR IN YEAR 1
0.0%
Year 1 is the one year an escalator cannot protect - only a higher starting price can.

MARGIN YOU RECOVER AT THE BREAKEVEN ESCALATOR
3.8%
How much final-year margin the escalator buys back.

How we got there

1. What you signed and what it costs

You bill $7.47M and it costs you $5.16M in year 1, leaving a 31.0% margin. Your floor is 12%, so there is 19.0% of room before this deal becomes unsignable.

Evidence: 19.0% of headroom

2. Why a fixed price is the problem

Costs do not sit still. The biggest mover is Cloud Hosting, which can swing 22.0% in a year, and the increases compound - year 3 carries three years of them. A frozen price carries none of that.

Evidence: 6.7% expected cost drift per year

3. Where it breaks

Across 1,000 complete cash flows, the Year 3 margin comes in below 12% in 18.4% of them, and at least one year breaks it in 18.5%. Year 3 is the weakest, with an expected margin of 20.3%. In today's money that is $961k of planned profit at risk.

Evidence: 18.4% chance of a breach in Year 3

4. What fixes it

A 2.5% annual escalator pulls the Year 3 risk back inside your 10% tolerance. Beyond the escalator, buy the right to reopen pricing: if, at the start of Year 3, delivered margin is below 12%, you renegotiate or re-scope.

Evidence: escalator 2.5%, trigger at Year 3

Charts

Margin over the life of the contract

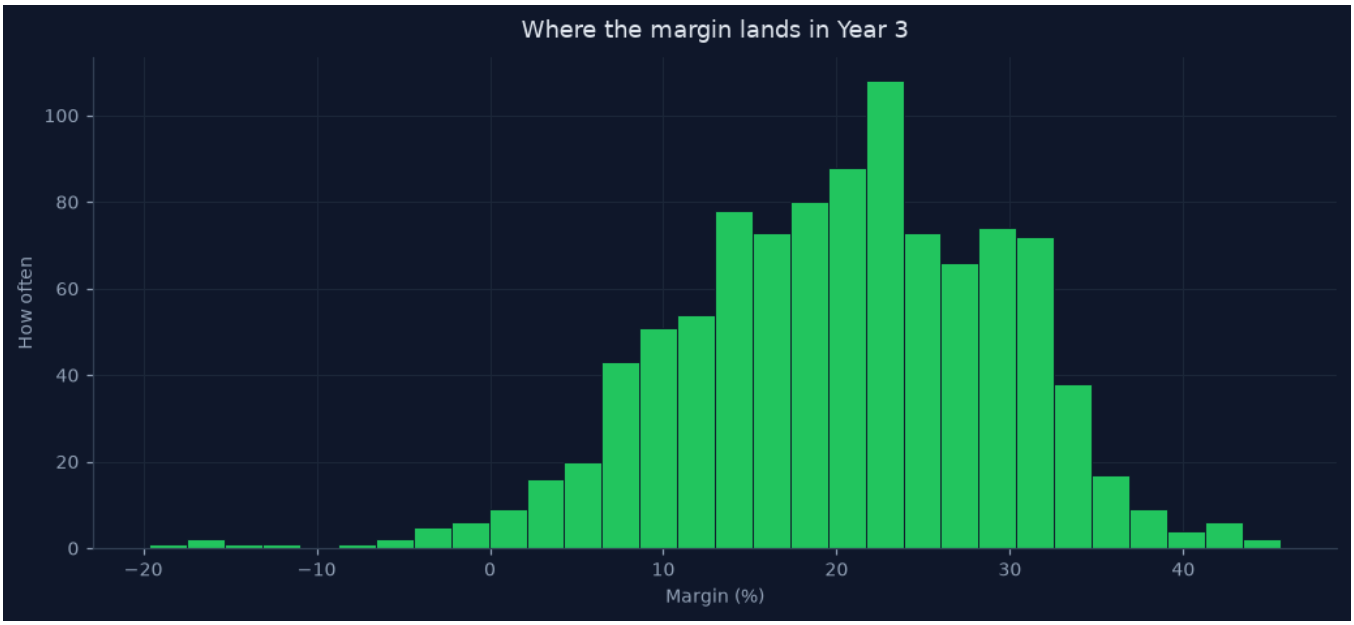
The middle line is the most likely path; the band is the range



What this shows: As long as the red line stays above the orange floor line you are safe. The moment it crosses, this contract can break your minimum margin rule.

Where the margin lands in Year 3

1,000 projections - the marker is your 12% floor



What this shows: Everything to the left of the floor marker is a world where this contract has broken your minimum margin rule.

Chance of breaking the floor, year by year

Your tolerance is 10% - anything above that bar is not signable as-is



What this shows: Risk builds as the contract ages because cost increases compound while a frozen price does not.

Tables

Year by year

Every one of the 1,000 projections, summarised one year at a time.

Year	You bill	Expected cost	Expected margin	A bad run (1 in 20)	A good run (1 in 20)	Chance below the floor	Chance of a loss
Year 1	$7,470,000	$5,157,039	31.0%	31.0%	31.0%	0.0%	0.0%
Year 2	$7,470,000	$5,539,994	25.8%	15.7%	35.6%	1.7%	0.1%
Year 3	$7,470,000	$5,953,906	20.3%	4.8%	33.8%	18.4%	1.9%

Row 3 is the year most likely to break your floor.

Searching for the breakeven escalator

The same 1,000 projections re-run with a different yearly price increase. We measure the risk in Year 3, the year the cost increases have compounded the longest.

Annual escalator	Chance of breaking the floor in Year 3	Chance of breaking it at some point	Expected margin in Year 3	Margin in today's money	Verdict
0.0%	18.4%	18.5%	20.3%	$5.00M	too risky
0.5%	16.5%	16.7%	21.1%	$5.09M	too risky
1.0%	14.5%	14.8%	21.9%	$5.18M	too risky
1.5%	12.2%	12.6%	22.6%	$5.28M	too risky
2.0%	10.3%	10.5%	23.4%	$5.37M	too risky
2.5%	8.4%	8.5%	24.1%	$5.46M	inside tolerance
3.0%	6.5%	6.6%	24.9%	$5.55M	inside tolerance
3.5%	5.8%	6.1%	25.6%	$5.65M	inside tolerance
4.0%	5.0%	5.5%	26.3%	$5.74M	inside tolerance
4.5%	4.2%	4.6%	27.0%	$5.83M	inside tolerance
5.0%	3.8%	4.2%	27.7%	$5.93M	inside tolerance
5.5%	2.8%	3.1%	28.4%	$6.02M	inside tolerance
6.0%	2.3%	2.6%	29.1%	$6.12M	inside tolerance
6.5%	1.9%	2.2%	29.7%	$6.21M	inside tolerance
7.0%	1.7%	2.0%	30.4%	$6.31M	inside tolerance
7.5%	1.6%	1.9%	31.0%	$6.40M	inside tolerance
8.0%	1.2%	1.5%	31.7%	$6.50M	inside tolerance

The highlighted row is the first one marked 'inside tolerance' - that is the escalator to ask for.

What goes into the model

Cost mix from 4 cost lines observed across 4 years in your file.

Cost line	Share of cost	Year 1 cost	Expected drift per year	Volatility	Share of the Year 3 wobble	Expected extra cost by Year 3
Cloud Hosting	30.3%	$1,562,709	9.6%	22.0%	47.7%	$360,611
Third-party Licensing	21.8%	$1,124,574	7.1%	12.0%	6.4%	$173,563
Labor	37.9%	$1,954,101	4.6%	9.0%	9.7%	$191,769
Logistics	10.0%	$515,655	5.0%	18.0%	2.9%	$62,502
Currency	-	-	0%	6.0%	33.3%	$10,712

The bottom line

What you planned versus what the simulation says, in today's money.

Column 1	Planned (margin held flat)	What the simulation says	Difference
Margin in Year 1	31.0%	31.0%	0.0%
Margin in Year 3	31.0%	20.3%	-10.7%
Margin in today's money	$5.96M	$5.00M	-$961k
Chance of breaking the floor in Year 3	0%	18.4%	18.4%
Chance of breaking it at some point	0%	18.5%	18.5%

Row 3 is the value at risk from cost volatility.

What we recommend

Do not sign a frozen price. Add a 2.5% annual escalator plus a renegotiation trigger at Year 3.

Risk Moderate 62 out of 100 | Confidence: High

Why

• Expected margin falls from 31.0% in Year 1 to 20.3% by Year 3.

• Chance of breaking the 12% floor by Year 3: 18.4% against a tolerance of 10.0% (18.5% if you count every year).

• Margin at risk, in today's money: $961k of the $5.96M you planned to book.

What to do

1. Add an annual cost-of-living escalator of 2.5% to the price - that alone brings the chance of breaking the floor in Year 3 from 18.4% to 8.4%.

2. Add a renegotiation trigger: if delivered margin is below 12% at the start of Year 3, you may reopen pricing or re-scope the service.

3. An escalator cannot protect Year 1 (it starts from Year 2). If the 0.0% Year-1 risk is not acceptable either, raise the starting price rather than relying on the escalator.

4. Cap or pass through the Cloud Hosting line first: it drives 47.7% of the cost uncertainty.

The trade-off: Every point of escalator you win protects about $384k of what that money is worth today - and every point you ask for is a point buyer will not wear the escalator, take the renegotiation trigger instead: it costs nothing upfront and gives you an exit.

What we assumed and what to watch

Worth knowing before you act

• With no escalator there is roughly a 1 in 5 chance of breaking your 12% floor by Year 3 - that is the whole risk of a frozen price.

• Your file shows Cloud Hosting, Third-party Licensing, Labor, Logistics moving too smoothly to measure a real swing from, so we kept your volatility setting for them rather than reading near-zero risk off a short history.

• Your file reports an average margin of 31.0%, so that - not the 15.0% you typed - sets year-1 cost. If you meant the figure you typed, clear the margin column from the mapping and run again.

• Contract value was read from your file ($7.47M average), overriding the $8.00M you typed.

Assumptions

• Contract: $7,470,000 billed per year for 3 years, starting margin 31.0%, so year-1 cost is $5,157,039.

• Cost mix: 4 cost lines observed across 4 years in your file - Cloud Hosting 30.3% of cost, drift 9.6% per year, volatility 22.0%, Third-party Licensing 21.8% of cost, drift 7.1% per year, volatility 12.0%, Labor 37.9% of cost, drift 4.6% per year, volatility 9.0%, Logistics 10.0% of cost, drift 5.0% per year, volatility 18.0%.

• Currency: a separate wobble of 6.0% applied to the whole cost base.

• Capacity cliff: none modelled (set the step jump above 0% to include one).

• Attrition: none modelled (set a backfill cost above 0% to include it).

• Each cost line follows a 'bad news weighs more than good news' curve: the middle outcome is the drift, but surprise increases are larger than surprise windfalls.

• Year 1 is priced on the cost you can see today; every year after that picks up a fresh cost shock over 2 year(s).

• Each volatility figure is read as the total swing for that cost line across the shocked years and spread evenly over them, so a 22.0% figure means 'ends up within about that much of plan by Year 3', not 'that much every single year'. Costs still compound - good and bad years stack up. Year-on-year volatility learned from your file is floored at 4%, because we will not assume a cost line is steadi…

• Cost lines in the same year move together somewhat (they share the same economy); years move independently of each other.

• Margin floor 12%, risk tolerance 10%, discount rate 8.0% used to value future money, escalator in this run 0.0%.

• 1,000 complete cash flows on an even grid; seed 42 - re-running gives identical numbers.

• Contract ID -> your 'contract_id' column

• Customer name -> your 'customer' column

• Contract value (revenue) -> your 'contract_value' column

• Year -> your 'year' column

• Cost component -> your 'cost_component' column

• Cost in that year -> your 'base_cost' column

• Margin achieved (%) -> your 'margin_pct' column

• Capacity cliff flag -> not in your file; not used

• Staff leavers (count) -> not in your file; not used

Appendix: the last steps of the run

Every line was produced while the model ran. Re-running with the same seed replays it exactly.

Time	Stage	What happened
0.00s	prepare	Cost component -> your 'cost_component' column
0.00s	prepare	Cost in that year -> your 'base_cost' column
0.00s	prepare	Margin achieved (%) -> your 'margin_pct' column
0.00s	prepare	Capacity cliff flag -> not in your file; not used
0.00s	prepare	Staff leavers (count) -> not in your file; not used
0.00s	prepare	Read 120 records into the model.
0.00s	simulate	Running the simulation…
0.00s	read	Reading the contract and how its cost is built up...
0.00s	read	Learned the cost mix and its drift from your file: 4 cost lines across 4 years.
0.00s	read	Your file shows 4 cost line(s) moving too smoothly to measure a swing from (Cloud Hosting, Third-party Licensing, Labor, Logistics), so we kept the volatility you typed in for them.
0.00s	read	Your file reports an average starting margin of 31.0%, so we used that rather than the default.
0.00s	read	Year 1: you bill $7.47M and it costs you $5.16M - a 31.0% margin.
0.00s	simulate	Laying out 1,000 complete 3-year cash flows across every possible path your costs could take...
0.01s	simulate	Walked 10% of the cash flows...
0.01s	simulate	Walked 20% of the cash flows...
0.01s	simulate	Walked 40% of the cash flows...
0.01s	simulate	Walked 50% of the cash flows...
0.01s	simulate	Walked 60% of the cash flows...
0.01s	simulate	Walked 70% of the cash flows...
0.01s	simulate	Walked 80% of the cash flows...
0.01s	simulate	Walked 90% of the cash flows...
0.01s	summarize	Reading the results year by year...
0.01s	summarize	Solving for the annual escalator that brings the risk back inside your tolerance...
0.01s	summarize	Found it: A 2.5% annual escalator brings the risk inside your 10% tolerance.

Time	Stage	What happened
0.01s	summarize	Done. 18.4% chance of breaking the floor by Year 3; you need a 2.5% annual escalator.